CNC Tool Management System

Software Requirements Gathering Questionnaire

This document contains a structured set of technical and functional questions designed to gather requirements for replacing your existing "Tooling" Google Sheet with a custom Next.js, React, Drizzle ORM, and Supabase web application.

Existing Sheet Structure Reference

Sheet Column	Target Data Type	Sample Data
`description`	String	"Harvey TopHat", "1mm drill", "6.7mm Gold-P Drill"
`shank`	String	"1/2", "3mm"
`shank (in)`	Decimal	0.5, 0.1181
`shank (m)`	Decimal	0.0127, 0.003
`collet (in)`	Decimal	0.5, 0.125
`diff in`	Decimal	0.0, 0.0069
`collet (m)`	Decimal	0.0127, 0.003175
`diff m`	Decimal	0.0, 0.000175
`best collet`	Decimal	0.5, 0.125
`cut dia`	Decimal	0.5, 0.0394
`radius`	Decimal / Null	0.015, *empty*
`LOC`	Decimal	0.75, 0.118
`series`	String	"HARV", "HSAM1"
`coating`	String / Boolean	"AlTiN", "True", "N"

Sheet Column	Target Data Type	Sample Data
`HSM`	Boolean / Null	"True", *empty*
`Tnum`	Integer	1, 12, 104
`Part`	String	"03-0085", "858632"
`Type`	String	"endmill", "drill", "ball", "special", "tap"

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1. Data Model & Database Schema (Drizzle + Supabase)

Identifiers & Keys

- Is `Tnum` (Tool Number) strictly unique across the entire library, or can tool numbers repeat across different tool types, pockets, or CNC machines? Tnum represents a unique cutting tool configuration (ie unique combination of tool type, diameter, length, etc). I may have multiple physical copies of a Tnum and i track my purchase history in cutterlog. But, in the magazine tab, a Tnum will be unique in the library, but, it's possible i have multiple copies of the same Tnum loaded into the cnc machine magazine. In otherwords, in the magazine tab under the listing of all tools, i only want one row for each Tnum.
- Should `Part` (Part Number/SKU) be unique, or do you stock multiple variations of the same part number (e.g., from different vendors or batches)? Part is not unique. I buy multiple copies of the same part number. Also, see above.
- Do you prefer a auto-generated UUID primary key behind the scenes for database integrity, while keeping `Tnum` as a user-facing unique field? Yes

Calculated Fields vs. Stored Data

- Imperial/Metric Conversions: Are values like `shank (m)` and `collet (m)` calculated dynamically from their inch equivalents (`shank (in) * 0.0254`), or do you need to store explicit, unrounded metric values separately? Tools are purchased in either metric or inch units. My machine is only using inch units. So, for tools that are natively metric i want to store the metric values and then always have the dynamic inch values available.
- Difference Columns: Are `diff in` and `diff m` strictly derived formulas (e.g., `collet - shank`), or can they be manually overridden by an operator? The should be derived formulas.
- Collet Selection: Is `best collet` driven by a strict mathematical rule (e.g., closest standard size exceeding shank diameter), or does it rely on a lookup table / human decision? Yes, it is closest standard size exceeding shank diameter. But, it needs to be able to choose between metric and inch collets.

Data Cleanliness & Nullability

- Should fields like `radius`, `HSM`, and `coating` allow null/empty states, or should they default to explicit baseline values (e.g., `radius = 0.0000`, `HSM = false`, `coating = "None"`)? They should allow null
- What fixed set of options should exist for `Type`? (e.g., `endmill`, `drill`, `ball`, `bull_nose`, `thread_mill`, `tap`, `special`). Should users be allowed to add custom tool types? Those are the correct fixed types and yes somewhere there should be an option to add new fixed types to the list.
- Is `coating` a simple flag (Yes/No), or a standard text string (e.g., "AlTiN", "ZrN", "Uncoated")? String.

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2. User Interface & Tool Manipulation

Views & Layout

- Do you prefer a compact inline-editable spreadsheet view (like AG-Grid or TanStack Table) where data can be edited directly in rows, or a master-detail view with a slide-over panel for editing tool properties? I want don't want an editable sheet. I want two different views/tabs. 1) a form where i enter new tools into the database. 2) a magazine which shows which tools are loaded in the cnc machine and next to it a list of all my tools. I can filter that list to show only assembled tools. And then i want to drag and drop between that list and what's in the magazine.
- What are the most frequent filter criteria you use during setup? (e.g., Filter by `Type`, show only `HSM` compatible tools, filter by `shank` size). I don't have an answer right now.
- Should search extend across all fields simultaneously (`description`, `Part`, `series`, `Tnum`), or do you need field-specific search inputs? Ok, for the magazine tab with the list of all tools, i think having filter options for each column would be best and maybe search with-in those columns.

Tool State & Operations

- What operational actions are involved in "manipulating tool settings"?
    - Updating tool assignment to specific CNC machine turrets/carousels? Right now just assembled or unassembled. Ideally I would also include what tool holder or collet is used with tool but that is future work.
    - Tracking tool wear, tool life, or offset adjustments (e.g., Z-offset, diameter wear)? No i'm not tracking that.
    - Marking tools as "Active", "In-Use", "Worn", "Broken", or "Out of Stock"? I may want to mark out of stock yes.
- Do you need an audit trail or history log showing who changed a tool's parameters and when? no.

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3. Autodesk Fusion Integration (Python Add-in)

Payload Handling & Job Lifecycle

- When the Python Add-in in Autodesk Fusion posts a JSON tool list to the Next.js API, what should happen immediately in the UI?
    - Should the app automatically filter or highlight the required tools for that CAM setup? I think in the magazine tab i would like a list of needed tools for the job, and then show whether they are current assembled and/or in the magazine
    - Should it prompt the user to create or assign a new "Job" / "Setup Sheet"? No, see above.
- How should missing tools be handled? If Fusion sends a tool with a `Tnum` or `Part` that does not exist in the Supabase database, should the app:
    - Automatically draft a new tool record based on the Fusion parameters? No.
    - Flag a blocking validation warning to the user? In the “magazine tab” i think i want the fusion output to be in like a separate square on the screen and if the tool coming from fusion doesn’t exist in the subabase database just highlight that in the separate square.

Data Synchronization

- Is the Fusion integration unidirectional (Fusion sends CAM setup tool lists to web app) or bidirectional (web app updates tool parameters back inside Fusion's Tool Library)? I would like to build in the bidirectional capability, but, right now it is only one way from fusion to the web application. Please include some foobar like test code for receiving info from the web application.
- Do you need to track live spindle/pocket assignments (e.g., Tool #12 is currently physically loaded into Machine A, Pocket 4)? Yes that will be tracked in the web app as noted above but i dont want that sent to fusion (for now).
- How should the application compare the Autodesk Fusion 360 output ("tools needed for current job") against the tool assemblies currently marked as loaded "in the magazine"? See above. In the separate “square” show the list of fusion specified tools and highlight if in the web app they are currently listed as assembled and in the machine. When i make changes in the web app update the “square”.
- How should the UI highlight missing tools or assemblies that need to be physically built, pre-set, and loaded for the incoming job? Yes! See above.

Magazine & Tool Assembly Tracking

Tool Assembly Structure & Workflow

- How do you define a "Tool Assembly"? Does it consist of specific components like a cutter, collet, pull stud, and tool holder? It consists of the Tnum (cutter), collet (if required) and toolholder type. Currently I am only tracking Tnum but you can include the capability for collets and toolholder type but keep it hidden for now.

- What workflow do you prefer for assembling and disassembling tools in the UI (e.g., drag-and-drop interface, modal dialogs, or multi-select dropdowns)? See above, drag and drop.

CNC Magazine State & Tool Movement

- How should the state of the CNC magazine be tracked (e.g., mapping which specific tool assembly is loaded into which magazine pocket/slot)? The cnc has Tnums installed to pockets. They are labelled as p-nums or pockets in the google sheet. So yes, map Tnnum to Pocket number. I only have one machine and it has 21 pockets.
- What process or interface action should operators use to move tool assemblies in and out of the machine's magazine quickly during setup? Drag and drop from the magazine list to the tool list in the magazine tab.

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4. Deployment, Multi-Mac Sync & Environment Setup

Authentication & Access

- The application will run across multiple Mac workstations. How should access be controlled?
    - **Option A:** Single Shared Access (No login required, network access only). This option A. I would like to use the same subabase credentials on each machine, like logging into gmail from different machines.
    - **Option B:** Simple Pin/Password (Shared team login). No
    - **Option C:** Supabase Auth with individual user accounts (e.g., Magic Link, Google Workspace, or Email/Password). No.
- Are all shop users permitted to edit and delete tools, or should operators have read-only views while programmers have full edit access? Yes. there are no access restrictions at all on the web app. Of course, i need supabase access to be restricted from the outside world.

Deployment Infrastructure

- Will the web app be hosted on a cloud platform (e.g., Vercel) connected to Supabase Cloud, or locally hosted on a Mac mini / local shop server? I would like it running on mac localhost. And querying supabase for data retrieval.
- Do you require offline capabilities on local Macs if internet connectivity drops, or is a persistent cloud connection acceptable? Since it’s local host i want offline capability except for the need to connect to supabase. If the internet goes out and i lose connection to supabase it’s ok for the web-app to be inoperable.
- Are environment secrets (Supabase API keys, database connection strings) managed via standard Vercel environment configurations and standard `.env.local` files for local development? Yes that sounds good unless you have a better idea.

Cutterlog & Purchase Tracking

Purchase Data & Inventory Integration

- What specific data points need to be recorded for purchases (e.g., order date, vendor, unit price, quantity, PO number)? You can include all those fields but allow them to be blank
- How should purchase history link back to the main tool inventory? Does logging a new purchase automatically increment a "quantity on hand" field for that specific Part/Tnum? I don’t need this functionality. What i want is a single database table for all the cutters. But, in the “magazine” tab i only want to show a single row for a tnum. So, if i have 5 copies of T3, in the magazine tab i only see one row for T3.